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SMMR
STATISTICAL METHODS IN MEDICAL RESEARCH

Statistical Methods in Medical Research
2018, Vol. 27(6) 1766–1784
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DOI: 10.1177/0962280216669182
journals.sagepub.com/home/smm
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ANOVA model for network meta-analysis of diagnostic test accuracy data

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Abstract

Procedures combining and summarising direct and indirect evidence from independent studies assessing the diagnostic accuracy of different tests for the same disease are referred to network meta-analysis. Network meta-analysis provides a unified inference framework and uses the data more efficiently. Nonetheless, handling the inherent correlation between sensitivity and specificity continues to be a statistical challenge. We developed an arm-based hierarchical model which expresses the logit transformed sensitivity and specificity as the sum of fixed effects for test, correlated study-effects to model the inherent correlation between sensitivity and specificity and a random error associated with various tests evaluated in a given study. We present the accuracy of 11 tests used to triage women with minor cervical lesions to detect cervical precancer. Finally, we compare the results with those from a contrast-based model which expresses the linear predictor as a contrast to a comparator test. The proposed arm-based model is more appealing than the contrast-based model since the former permits more straightforward interpretation of the parameters, makes use of all available data yielding shorter credible intervals, and models more natural variance–covariance matrix structures.

Keywords

Meta-analysis, network meta-analysis, diagnostic tests, hierarchical model, arm-based

1 Introduction

Network meta-analyses (NMA) have classically been used to extend conventional pairwise meta-analyses by combining and summarising direct and indirect evidence on multiple 'therapeutic' interventions for a given condition when the set of evaluated interventions/treatments differs among studies. By borrowing strength from the indirect evidence, there is a potential gain in precision of the estimates.[1] Furthermore, the estimates may be less biased and more robust. Such an approach uses the data efficiently and is line with the principle of intention-to-treat (ITT)[2] in randomised clinical trials which requires that all valid available data should be used even when a part of the data is missing. In a diagnostic test accuracy study, an index test and possibly one or more comparator tests are administered to each tested subject. A standard or reference test or procedure is also applied to all the patients to classify them as having the target condition or not. The patients' results are then categorised by the index and reference test as true positive, false positive, true negative and false negative.

The diagnostic accuracy of the index test is represented as a bivariate outcome and is typically expressed as sensitivity and specificity at a defined test cut-off. Differences due to chance, design, conduct, patients/participants, interventions, tests and reference test imply that there will be heterogeneity often in opposite direction for the two typical accuracy outcomes: sensitivity and specificity. While traditional meta-analyses allow for comparison between two tests, there are often multiple tests for the diagnosis of a particular disease outcome. To present the overall picture, inference about all the tests for the same condition and patient characteristics is therefore required. The simultaneous analysis of the variability in the accuracy of multiple tests within and between studies may be approached through a network meta-analysis.

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In combining univariate summaries from studies where the set of tests differs among studies, two types of linear mixed models have been proposed. The majority of NMA express treatment effects in each study as a contrast relative to a baseline treatment in the respective study.[1,3] This is the so-called contrast-based (CB) model. Inspired by the CB models developed for interventional studies, Menten and Lesaffre[4] introduced a CB model for diagnostic test accuracy data to estimate the average log odds ratio for sensitivity and specificity of the index test relative to a baseline or comparator test.

The second type of models is the classical two-way ANOVA model with random effects for study and fixed effect for tests,[5–7] the so-called arm-based (AB) model. The AB model is based on the assumption that the missing arms or tests are missing at random. While the two types of models yield similar results for the contrasts with restricted maximum likelihood (REML) procedures, the CB model is generally not invariant to changes in the baseline test in a subset of studies. The CB model yields odds ratios (OR) making it difficult to recover information on the absolute diagnostic accuracy (the marginal means), relative sensitivity or specificity of a test compared to another or differences in accuracy between tests: measures that are easily interpretable and often used in clinical epidemiology. It is common knowledge that the OR is only a good approximation of relative sensitivity/specificity when the outcome is rare,[8–10] but this is often not the case in diagnostic studies. Moreover, the AB model is simpler when the baseline/comparator treatment varies from one study to another or when the number of tests varies substantially among studies. By accommodating more complex variance–covariance structures, AB models have been shown to be superior to CB models.[11]

We apply the two-way ANOVA model in a diagnostic data setting by extending the AB model in two ways: (1) using two independent binomial distributions to describe the distribution of true positives and true negatives among the diseased and the healthy individuals, and (2) inducing a correlation between sensitivity and specificity by introducing correlated and shared study effects. The resulting generalised linear mixed model is analogous to randomised trials with complete block designs or repeated measures in analysis of variance models where studies are equivalent to blocks. The main assumption is that, results missing for some tests and studies are missing at random. This approach is efficient because the correlation structure allows the model to borrow information from the 'imputed' missing data to obtain adjusted sensitivity and specificity estimates for all the tests.

2 Motivating dataset

To illustrate the use of the proposed model in network meta-analysis of diagnostic test accuracy data, we analyse data on a diversity of cytological or molecular tests to triage women with equivocal or mildly abnormal cervical cells.[12–16] A Pap smear is a screening test used to detect cervical precancer. When abnormalities in the Pap smear are not high grade, a triage test is needed to identify the women who need referral for further diagnostic work-up. There are several triage options, such as repetition of the Pap smear or HPV DNA or RNA assays. HPV is the virus causing cervical cancer.[17] Several other markers can be used for triage as well, such as p16 or the combinations of p16/Ki67 which are protein markers indicative for a transforming HPV infection.[15,18]

The data are derived from a comprehensive series of meta-analyses on the accuracy of triage with HPV assays, cervical cytology or molecular markers applied on cervical specimens in women with minor cervical abnormalities.[12–16] Two patient groups with minor cytological abnormalities were distinguished: women with ASC-US (atypical squamous cells of unspecified significance) and women with LSIL (low-grade squamous intraepithelial lesions). Two levels of precancer (disease) were considered: intraepithelial neoplasia lesion of grade 2 or worse (CIN2+) or of grade 3 or worse (CIN3+). The disease status was ascertained by colposcopy. This was followed by a partial histological verification of a biopsy specimen when colposcopy was positive.

In total, the accuracy of 11 tests for detecting cervical precancer was evaluated. Labelled 1–11, the tests were hrHPV DNA testing with Hybrid Capture II (HC2), conventional cytology (CC), liquid-based cytology (LBC), generic PCRs targeting hrHPV DNA (PCR) and commercially available PCR-based hrHPV DNA assays such as: Abbott RT PCR hrHPV, Linear Array and Cobas-4800; assays detecting mRNA transcripts of 5 (HPV Proofer) or 14 (APTIMA) HPV types; and protein markers identified by cytoimmunochestry such as p16 and p16/Ki67, which are over-expressed as a consequence of HPV infection. A total of 125 studies with at least one test and maximum of six tests were included allowing assessment of the accuracy of the 11 triage tests.

The size of the nodes in Figure 1 is proportional to the number of studies evaluating a test, and thickness of the lines between the nodes is proportional to the number of direct comparisons between tests. The size of the node and the amount of information in a node consequently influence the standard errors of the marginal means and the

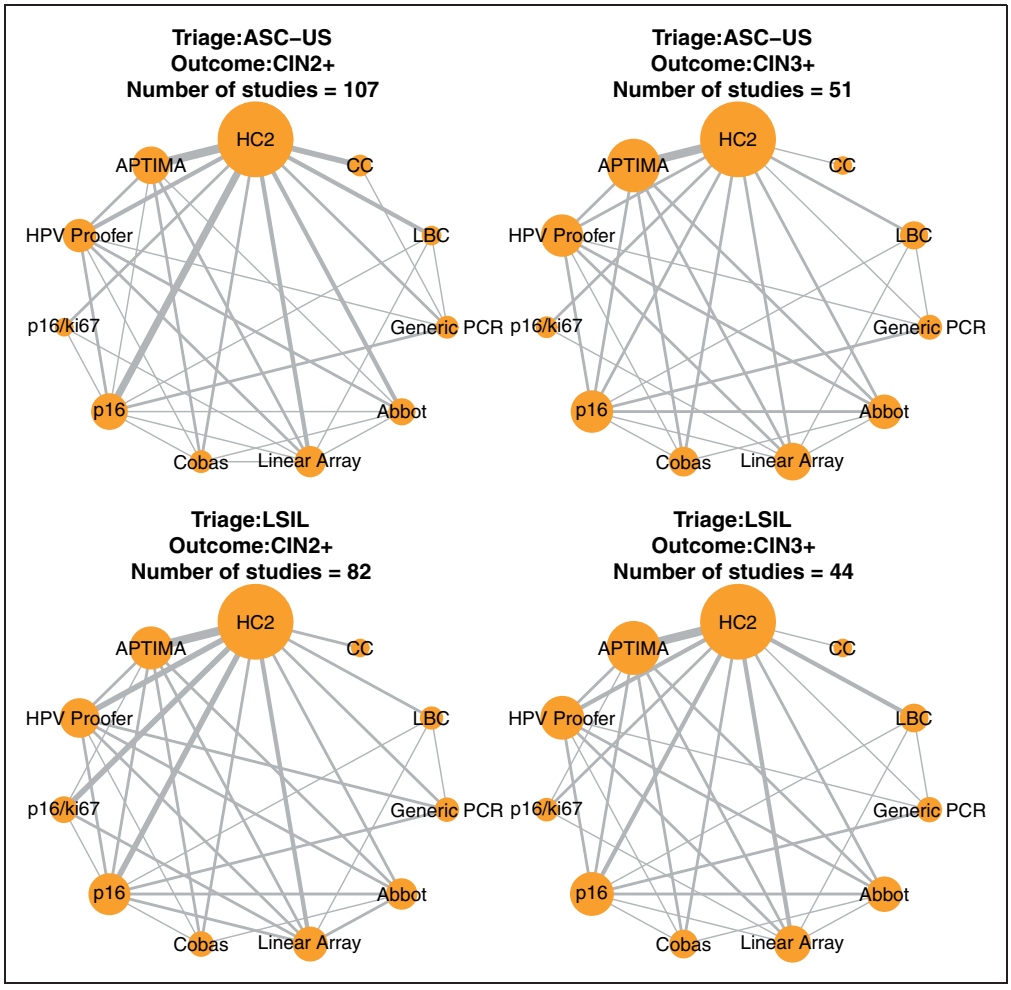


Figure 1. Network plot of all included tests to triage women with atypical squamous cells of unspecified significance (ASC-US) and low-grade squamous intraepithelial lesions (LSIL) to detect intraepithelial neoplasia lesion of grade 2 or worse (CIN2+) or of grade 3 or worse (CIN3+). The size of the nodes and connecting lines are proportional to the number of evaluations of test and number of test comparisons, respectively. HC2 and APTIMA were the most commonly assessed tests.

relative measures. From the network plot, HC2 and APTIMA were the most commonly assessed tests. The network in Figure 1 is connected because there exists at least one study evaluating a given test together with at least one of the other remaining 10 tests.

3 Methodology

Suppose there are K tests and I studies. Studies assessing two tests ($k = 2$) are called 'two-arm' studies while those with $k > 2$ are 'multi-arm' studies. For a certain study i, let (Y_{i1k}, Y_{i2k}) denote the true positives and true negatives, (N_{i1k}, N_{i2k}) the diseased and healthy individuals and (π_{i1k}, π_{i2k}) the 'unobserved' sensitivity and specificity, respectively, with test k in study i. Given study-specific sensitivity and specificity, two independent binomial distributions describe the distribution of true positives and true negatives among the diseased and the healthy individuals as follows

$$Y_{ijk} \mid \pi_{ijk}, x_i \sim bin(\pi_{ijk}, N_{ijk}), \quad i = 1, \ldots I, j = 1, 2, k = 1, \ldots K \tag{1}$$

where x_i generically denotes one or more covariates, possibly affecting π_{ijk}. In the next section, we present the recently introduced contrast-based model[4] followed by our proposed arm-based model to estimate the mean as well as comparative measures of sensitivity and specificity.

3.1 Contrast-based model

By taking diagnostic test T_K as the baseline, Menten and Lessafre[4] proposed the following model

$$\begin{aligned}
logit(\pi_{ijk}) &= \theta_{ijk} \\
\theta_{ij1} &= \mu_{ij} + (K-1)\times\frac{\delta_{ij1}}{K} - \frac{\delta_{ij2}}{K} - \frac{\delta_{ij3}}{K} - \ldots - \frac{\delta_{ij(k-1)}}{K} \\
\theta_{ij2} &= \mu_{ij} - \frac{\delta_{ij1}}{K} + (K-1)\times\frac{\delta_{ij2}}{K} - \frac{\delta_{ij3}}{K} - \ldots - \frac{\delta_{ij(k-1)}}{K} \\
&\vdots \\
\theta_{ijK} &= \mu_{ij} - \frac{\delta_{ij1}}{K} - \frac{\delta_{ij2}}{K} - \frac{\delta_{ij3}}{K} - \ldots - \frac{\delta_{ij(k-1)}}{K} \\
\boldsymbol{\delta_i} &= (\delta_{i11},\ldots\ \delta_{i1(K-1)}, \delta_{i21},\ \ldots, \delta_{i2(K-1)}) \ \sim\ N(\boldsymbol{\nu_\delta}, \boldsymbol{\Sigma}) \\
\boldsymbol{\nu_\delta} &= (\nu_{\delta 11}, \ldots, \nu_{\delta 1(K-1)}, \nu_{\delta 21}, \ldots, \nu_{\delta 2(K-1)})
\end{aligned} \tag{2}$$

The $\boldsymbol{\nu_\delta}$ represents the average log odds ratio for sensitivity and specificity of the $K-1$ tests compared to the baseline test T_K.

The matrix $\boldsymbol{\Sigma}$ models the variances and co-variances of the contrasts $\boldsymbol{\delta_i}$ and contributes in a complex manner to the variances and co-variances of $\boldsymbol{\theta_i}$. In fact, it would be almost futile to express the corresponding matrix $\boldsymbol{\Sigma}_{\boldsymbol{\theta_i}}$.

There are known difficulties in estimating the variance–covariance matrix $\boldsymbol{\Sigma}$ since each sampled matrix should be positive-definite.[19] Moreover, the components of $\boldsymbol{\Sigma}$ are difficult to comprehend and depend on baseline test. Furthermore, the model identification becomes difficult as the number of tests included increases.

To enable model identification, the authors therefore recommend a diagonal or block diagonal variance–covariance matrix $\boldsymbol{\Sigma}$ expressed as follows

$$\boldsymbol{\Sigma}_{\boldsymbol{diagonal}} = \begin{bmatrix}
\sigma^2_{\delta_{i11}} & \ldots & 0 & 0 & \ldots & 0 \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & \ldots & \sigma^2_{\delta_{i1K-1}} & 0 & \ldots & 0 \\
0 & \ldots & 0 & \sigma^2_{\delta_{i21}} & \ldots & 0 \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & \ldots & 0 & 0 & \ldots & \sigma^2_{\delta_{i2K-1}}
\end{bmatrix} \tag{3}$$

or

$$\boldsymbol{\Sigma}_{\boldsymbol{block\ diagonal}} = \begin{bmatrix}
\sigma^2_{\delta_{i11}} & \ldots & \sigma_{\delta_{i11},\delta_{i1(K-1)}} & 0 & \ldots & 0 \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
\sigma_{\delta_{i11},\delta_{i1(K-1)}} & \ldots & \sigma^2_{\delta_{i1K-1}} & 0 & \ldots & 0 \\
0 & \ldots & 0 & \sigma^2_{\delta_{i21}} & \ldots & \sigma_{\delta_{i21},\delta_{i2(K-1)}} \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
0 & \ldots & 0 & \sigma_{\delta_{i21},\delta_{i2(K-1)}} & \ldots & \sigma^2_{\delta_{i2K-1}}
\end{bmatrix} \tag{4}$$

While this reduces model complexity and difficulty in estimation, such covariance matrices totally ignore the correlation between the logit sensitivity and logit specificity.

The authors estimate the absolute accuracy of the tests from the estimated $logit^{-1}(\mu_{jk})$ as follows

$$\begin{aligned}
E(\widehat{\mu}_j) &= \frac{1}{I}\sum_{i=1}^{I}\mu_{ij} \\
\mu_{j1} &= logit^{-1}(E(\widehat{\mu}_j)) + (K-1)\times\frac{\nu_{\delta j1}}{K} - \frac{\nu_{\delta j2}}{K}s - \frac{\nu_{\delta j3}}{K}\ldots - \frac{\nu_{\delta j(K-1)}}{K}
\end{aligned}$$

$$\begin{aligned}
\mu_{j2} &= logit^{-1}(E(\widehat{\mu_j})) \; - \; \frac{\nu_{\delta j1}}{K} \; + \; (K-1) \times \frac{\nu_{\delta j2}}{K} \; - \; \frac{\nu_{\delta j3}}{K} \ldots \; - \; \frac{\nu_{\delta j(K-1)}}{K} \\
&\vdots \\
\mu_{jK} \; &= \; logit^{-1}(E(\widehat{\mu_j})) \; - \; \frac{\nu_{\delta j1}}{K} \; - \; \frac{\nu_{\delta j2}}{K} \; - \; \frac{\nu_{\delta j3}}{K} \ldots \; - \; \frac{\nu_{\delta j(K-1)}}{K}
\end{aligned} \tag{5}$$

where $logit^{-1}(E(\widehat{\mu_j}))$ is the average probability of testing positive/negative. Equation (5) estimates the accuracy of tests for a hypothetical study with random-effects equal to zero but not the meta-analytic estimates. The meta-analytic estimates are obtained after the following complex integration

$$\begin{aligned}
E(\pi_{ij1}) &= \int logit^{-1}\left(\mu_{ij} \; + \; (K-1) \times \frac{\delta_{ij1}}{K} - \frac{\delta_{ij2}}{K} \ldots - \frac{\delta_{ij(K-1)}}{K}\right) f(\boldsymbol{\delta_{ij}}) \mathrm{d}\delta_{ij1} \ldots \mathrm{d}\delta_{ij(K-1)} \\
E(\pi_{ij2}) &= \int logit^{-1}\left(\mu_{ij} \; - \; \frac{\delta_{ij1}}{K} + (K-1) \times \frac{\delta_{ij2}}{K} \ldots - \frac{\delta_{ij(K-1)}}{K}\right) f(\boldsymbol{\delta_{ij}}) \mathrm{d}\delta_{ij1} \ldots \mathrm{d}\delta_{ij(K-1)} \\
&\vdots \\
E(\pi_{ijK}) &= \int logit^{-1}\left(\mu_{ij} \; - \; \frac{\delta_{ij1}}{K} - \frac{\delta_{ij2}}{K} \ldots - \frac{\delta_{ij(K-1)}}{K}\right) f(\boldsymbol{\delta_{ij}}) \mathrm{d}\delta_{ij1} \ldots \mathrm{d}\delta_{ij(K-1)}
\end{aligned} \tag{6}$$

Using MCMC methods, the meta-analytic estimates can be obtained as follows

$$\widehat{E(\pi_{ijk})} = \frac{1}{I} \sum_{i=1}^{I} logit^{-1}(\theta_{ijk}) \tag{7}$$

3.2 Arm-based model

Consider a design where there is at least one test per study. The study serves as a block where all diagnostic accuracy tests are hypothetically evaluated of which some are missing. This modelling approach has potential gain in precision by borrowing strength from studies with single tests as well as multi-arm studies. The proposed single-factor design with repeated measures model is written as follows

$$\begin{aligned}
logit(\pi_{ijk}) &= \theta_{ijk} = \mu_{jk} + \eta_{ij} + \delta_{ijk} \\
\begin{pmatrix} \eta_{i1} \\ \eta_{i2} \end{pmatrix} &\sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \boldsymbol{\Sigma}\right) \\
\boldsymbol{\Sigma} &= \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \\
\boldsymbol{\delta_i} &= (\delta_{ij1}, \delta_{ij1}, \ldots \delta_{ijK}) \; \sim N(\mathbf{0}, diag(\boldsymbol{\tau_j^2}))
\end{aligned} \tag{8}$$

where $logit^{-1}(\mu_{1k})$ and $logit^{-1}(\mu_{2k})$ are the mean sensitivity and specificity in a hypothetical study with random-effects equal to zero, respectively. η_{ij} is the study effect for healthy ($j=1$) or diseased individuals ($j=2$) and represents the deviation of a particular study i from the mean sensitivity ($j=1$) or specificity ($j=2$), inducing between-study correlation. The study effects are assumed to be a random sample from a population of such effects. The common between-study variability of sensitivity and specificity and the correlation thereof are captured by the parameters σ_1^2, σ_2^2 and ρ, respectively. δ_{ijk} is the error associated with the sensitivity ($j=1$) or specificity ($j=2$) of test k in the ith study. Conditional on study i, the repeated measurements are independent with variance constant across studies such that $\boldsymbol{\tau_j^2} = (\tau_{j1}^2, \ldots, \tau_{jK}^2)$ is a K-dimensional vector of variances. The full variance–covariance

matrix of $\boldsymbol{\theta_i}$ is expressed as follows

$$\begin{bmatrix} \sigma_1^2+\tau_{11}^2 & \sigma_1^2 & \ldots & \sigma_1^2 & \rho\sigma_1\sigma_2 & \ldots & \ldots & \rho\sigma_1\sigma_2 \\ \sigma_1^2 & \sigma_1^2+\tau_{12}^2 & \ldots & \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \sigma_1^2 & \vdots & \ddots & \ddots & \vdots \\ \sigma_1^2 & \ldots & \sigma_1^2 & \sigma_1^2+\tau_{1K}^2 & \rho\sigma_1\sigma_2 & \ldots & \ldots & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \ldots & \ldots & \rho\sigma_1\sigma_2 & \sigma_2^2+\tau_{21}^2 & \sigma_2^2 & \ldots & \sigma_2^2 \\ \vdots & \ddots & \ddots & \vdots & \sigma_2^2 & \sigma_2^2+\tau_{22}^2 & \ldots & \vdots \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots & \ddots & \sigma_2^2 \\ \rho\sigma_1\sigma_2 & \ldots & \ldots & \rho\sigma_1\sigma_2 & \sigma_2^2 & \ldots & \sigma_2^2 & \sigma_2^2+\tau_{2K}^2 \end{bmatrix}$$

The correlation $\text{corr}(\theta_{ijk}, \theta_{ijk'})$ between kth and k'^{th} logit sensitivity ($j=1$) or specificity ($j=2$) is $\rho_{jkk'} = \frac{\sigma_j^2}{\sqrt{(\sigma_1^2+\tau_{jk}^2)\times(\sigma_1^2+\tau_{jk'}^2)}}$. In case $\tau_{j1}^2 = \ldots = \tau_{jK}^2 = \tau_j^2$ (homogeneous variances across tests), the shared random element η_{ij} within study i induces a non-negative correlation between any two test results k and k' from healthy individuals ($j=1$) or from diseased individuals ($j=2$) equal to $\rho_j = \frac{\sigma_j^2}{\sigma_j^2+\tau_j^2}$ (implying that a covariance matrix with compound symmetry). While it might seem logical to expect and allow for similar correlation between any two sensitivities or specificities in a given study, the variances τ_{jk}^2 of different sensitivities or specificities of the same study may be different. In such instances, the unstructured covariance matrix is more appropriate as it allows varying variances between the tests (in which case $\boldsymbol{\tau_j^2}$ is a K dimensional vector of the unequal variances).

ρ_j or ρ'_{jkk} is called the intra-study correlation coefficient which also measures the proportion of the variability in θ_{ijk} that is accounted for by the between-study variability. It takes the value 0 when $\sigma_j^2 = 0$ (if study effects convey no information) and values close to 1 when σ_j^2 is large relative to τ_j^2 and the studies are essentially all identical. When all components of $\boldsymbol{\tau_j^2}$ are equal to zero, the model reduces to fitting separate bivariate random-effect meta-analysis[20,21] model for each test.

On the other hand, the correlation between logit sensitivity ($j=1$) and specificity ($j=2$) of kth and k'^{th} ($k=k'$) or ($k \neq k'$) test equals

$$\rho_{12kk'} = \frac{\rho\sigma_1\sigma_2}{\sqrt{(\sigma_1^2 + \tau_{jk}^2) \times (\sigma_1^2 + \tau_{jk'}^2)}} \tag{9}$$

Homogeneous variances across tests imply that logit sensitivity and specificity are correlated in the same way in all the tests. This assumption may be reasonable if there is a common 'threshold' effect in all the cases. For network meta-analysis, different types of tests with different 'thresholds' are evaluated and the assumption of common correlation between sensitivity and specificity in all tests may no longer hold.

In most practical situations, the mean structure is of primary interest and not the covariance structure. Nonetheless, an appropriate covariance modelling is critical in the interpretation of the random variation in the data as well as obtaining valid model-based inference for the mean structure. Compound symmetry assumes homogeneity of variance and covariance and such restriction could invalidate inference for the mean structure when the assumed covariance structure is misspecified.[22]

When the primary objective of the analysis is on estimating the marginal means of sensitivity and specificity, the choice between compound symmetry and unstructured covariance structure is not critical because the inference procedure for the marginal means is the same. Moreover, over-parameterisation of the covariance structure might lead to inefficient estimation and potentially poor assessment of standard errors of the marginal means.[23] Watanabe-Akaike Information Criterion (WAIC), a measure of predictive accuracy of a fitted model, can be used to choose the appropriate covariance structure.[24]

In essence, the model separates the variation in the studies into two components: the within-study variation $diag(\boldsymbol{\tau_j^2})$ referring to the variation in the repeated sampling of the study results if they were replicated, and the between-study variation $\boldsymbol{\Sigma}$ referring to variation in the studies true underlying effects.

The study-level covariate information is included in the linear predictor in Equation 8 as follows

$$\theta_{ijk} = \sum_{p=1}^{P} X_{pi}\beta_{pjk} + \eta_{ij} + \delta_{ijk} \tag{10}$$

where β_{pjk} is the pth coefficient corresponding to the X_{pi} covariate in a hypothetical study with random effects equal to zero, respectively.

The population-averaged or the marginal sensitivity/specificity in the intercept-only model for test k is estimated as

$$\begin{aligned} E(\pi_{ijk}) &= E(logit^{-1}(\mu_{jk} + \eta_{ij} + \delta_{ijk})) \\ &= \int_{-\infty}^{\infty} logit^{-1}(\mu_{jk} + \eta_{ij} + \delta_{ijk}) f(\eta_{ij}) f(\delta_{ijk}) \, \mathrm{d}\eta_{ij} \, \mathrm{d}\delta_{ijk} \\ \widehat{E(\pi_{ijk})} &= \frac{1}{I}\sum_{1}^{I} logit^{-1}(\mu_{jk} + \eta_{ij} + \delta_{ijk}) \end{aligned} \tag{11}$$

The relative sensitivity and specificity and other relative measures of test k (relative to test $k', k \neq k'$) are then estimated from the marginal sensitivity or specificity of test k and k'.

3.3 Ranking of the tests

While ranking of tests using rank probabilities and rankograms is an attractive feature of univariate NMA, it is still a challenge to rank competing diagnostic tests especially when a test does not outperform the others on both sensitivity and specificity. Consider the diagnostic odds ratio (DOR)[25] which is expressed in terms of sensitivity and specificity as

$$DOR_k = \frac{sensitivity_k \times specificity_k}{(1 - sensitivity_k) \times (1 - specificity_k)} \tag{12}$$

and ranges from 0 to ∞ with: $DOR_k > 1$ or higher indicating better discriminatory test performance, $DOR_k = 1$ indicating a test that does not discriminate between the healthy and diseased, and $DOR_k < 1$ indicating an improper test. The DOR is a single indicator combining information about sensitivity and specificity and is invariant of disease prevalence. However, the measure cannot distinguish between tests with high sensitivity but low specificity or vice-versa.

Alternatively, the superiority of a diagnostic test could be quantified using a superiority index introduced by Deutsch et al.[26] expressed as

$$S_k = \frac{2a_k + c_k}{2b_k + c_k} \tag{13}$$

where a_k is the number of tests to which test k is superior (higher sensitivity and specificity), b_k is the number of tests to which test k is inferior (lower sensitivity and specificity) and c_k the number of tests with equal performance as test k (equal sensitivity and specificity). S ranges from 0 to ∞ with S tending to ∞ and S tending to 0 as the number of tests to which test k is superior and inferior increases, respectively, and S tending to 1 the more the tests are equal. Since the number of tests not comparable to test k does not enter into the calculation of S, the index for different tests may be based on different sets of tests.

3.4 Missing data and exchangeability

In the models above, not all the studies provide estimates of all effects of interest because some of the components of the vector $\mathbf{Y}_{ij} = (Y_{ij1}, \ldots, Y_{ijK})$ are missing. The $\mathbf{Y}_{ij}$ vector can be partitioned into the observed $\mathbf{Y}_{ij}^{o}$ and the missing $\mathbf{Y}_{ij}^{m}$. For each component of $\mathbf{Y}_{ij}$, let $\mathbf{R}_{ij}$ denote a vector of missingness indicator with

$$R_{ijk} = \begin{cases} 1 & \textit{if } Y_{ijk} \textit{ is observed,} \\ 0 & \textit{otherwise.} \end{cases}$$

The joint distribution of (**Y**, **R**), given the parameters (β, ϕ), is given by

$$p(y_{ij}, R_{ij}|\boldsymbol{\beta}_j, \phi_j) \tag{14}$$

where ϕ_j contains the missingness parameters and $\boldsymbol{\beta}_j$ contains $(\pi_{ij}, \boldsymbol{\Sigma}, \rho, \sigma_j, diag(\boldsymbol{\tau}_j))$. In a selection framework,[27,28] the joint distribution in equation (14) is factorised as

$$p(y_{ij} \mid \boldsymbol{\beta}_j), \phi_j \, p(R_{ij} \mid Y_{ij}, \phi_j) = p(y^o_{ij}, y^m_{ij} \mid \boldsymbol{\beta}_j, \phi_j) \, p(R_{ij} \mid Y^o_{ij}, Y^m_{ij}, \phi_j) \tag{15}$$

where $p(R_{ij} \mid y^o_{ij}, y^m_{ij}, \phi_j)$ describes the mechanism for data missingness. Assuming that the probability of missingness is conditionally independent of the unobserved data given the observed (so-called missing at random (MAR)), the second part of equation (15) simplifies to

$$p(R_{ij} \mid y^o_{ij}, y^m_{ij}, \phi_j) = p(R_{ij} \mid y^o_{ij}, \phi_j) \tag{16}$$

When the parameters β_{ij} and ϕ_{ij} are distinct and functionally independent, the missing data mechanism is ignorable and expression (16) can be dropped from the joint distribution in equation (15). Intergrating over the unknown missing values in the first part of equation (15) yields a marginal density with the observed information which is to be evaluated

$$\int p(y^o_{ij}, y^m_{ij} \mid \boldsymbol{\beta}_j) \, dy^m_{ij} = p(y^o_{ij} \mid \boldsymbol{\beta}_j) \tag{17}$$

Since the main objective is to be able to make valid and efficient inference about the parameters of interest and not to estimate or predict the missing data, the ignorability condition validates inference based on the observed data likelihood only. Conditional on π_{ijk}, the studies are assumed to be exchangeable. The observed information Y_{ijk} on a given test/arm k generically represents a point estimate of π_{ijk} and contributes to the estimation of the fixed effects μ_{jk}. At the second level of the hierarchy (equation (8)), exchangeable normal prior distributions with mean zero split the variability into between- and within-study variability. The observed data in each study contribute to the estimation of η_{ij} while all the studies all together contribute to the estimation of δ_{ijk} where δ_{ijk} and η_{ij} are considered independent samples from a population controlled by the hyper-parameters $\boldsymbol{\Sigma}$ and $\boldsymbol{\tau}^2_j$ which are estimated from the observed data. The hyper-parameters also have exchangeable vague or non-informative prior distributions. In fact, the exchangeability assumption is applied in both the CB and the AB models but in a different manner. The CB model assumes exchangeability of tests contrasts (odds ratios) across the studies while the AB assumes exchangeability of tests effects (means) across the studies.

3.5 Prior distributions

We decompose the covariance matrix $\boldsymbol{\Sigma}$ into a variance and correlation matrix such that

$$\boldsymbol{\Sigma} = diag(\sigma_1, \sigma_2) \times \boldsymbol{\Omega} \times diag(\sigma_1, \sigma_2) \tag{18}$$

where

$$\boldsymbol{\Omega} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

The model is completed by specifying vague priors on the mean, variance and correlation parameters as follows

$$\begin{aligned} tanh^{-1}(\rho), \mu_{jk} &\sim N(0, 25) \\ \tau_j, \sigma_j &\sim U(0, 5) \end{aligned} \tag{19}$$

Since it is not clear when certain choices of prior distributions are vague and non-informative, it is necessary to vary the prior distribution and assess their influence on the parameter estimates. The following prior distributions

were also used as part of a sensitivity analysis

$$\begin{aligned} \rho &\sim U(-1, 1) \\ \tau_j, \sigma_j &\sim cauchy(0, 2.5) \end{aligned} \tag{20}$$

An alternative prior distribution for the correlation matrix $\mathbf{\Omega}$ is the LKJ prior distribution with shape parameter $\nu = 1$ or $\nu = 2$[29]

$$\mathbf{\Omega} \sim LKJcorr(\nu) \propto det(\mathbf{\Omega})^{\nu-1} \quad for\ \nu \geq 1 \tag{21}$$

where ν controls the expected correlation with larger values favouring less correlation and vice-versa. Other possible prior distributions for $\mathbf{\Sigma}$ are: the Inverse-Wishart distribution having the advantage of computational convenience but being difficult to interpret or the more relaxed scaled inverse-Wishart which is a conjugate to the multivariate normal making Gibbs sampling simpler.[30]

3.6 Implementation

The models are fitted in the Bayesian framework using Stan,[31] a probabilistic programming language which has implemented Hamilton Monte Carlo (MHC) and No-U-Turn sampler (NUTS)[32] within R 3.3.0[33] using the rstan 2.9.0 package.[34] The Stan code for the model is provided alongside the supplementary material. We run three chains in parallel until there is convergence. Trace plots are used to visually check whether the distributions of the three simulated chains mix properly and are stationary.

For each parameter, convergence is assessed by examining the potential scale reduction factor $\hat{R}$, the effective number of independent simulation draws (n_{eff}), and the MCMC error. It is common practice to run simulations until $\hat{R}$ is no greater than 1.1 for all the parameters. Since Markov chain simulations tend to be autocorrelated, n_{eff} is usually smaller compared to the total number of draws. To reduce autocorrelation and consequently increase n_{eff}, it is necessary to do thinning by keeping every nth (e.g. every $10^{th}, 20^{th}, 30^{th}\ldots$) draw and discarding the rest of the samples. Besides, thinning saves memory especially when the total number of iterations is large.

4 Results

Figure 2 presents the study-specific sensitivity and specificity of all the 11 tests used to detect CIN2+in ASC-US triage from all available studies and from studies that evaluated at least two tests, one of them being HC2. We successively present the sensitivity and specificity of the 11 tests in triage of ASC-US and LSIL for outcomes CIN2+ and CIN3+, in Figures 3 to 5, respectively.

Representing the pooled sensitivity and specificity, the black diamonds are estimated by the AB model from all the available studies, the red diamond by the same model but from studies with at least two tests with one of them being HC2 while the blue diamonds are estimated by the CB model from studies with at least two tests with one of them being HC2.

Represented by the diamonds, the median was used as the measure of central tendency due to the asymmetry of sensitivity and specificity. The vertical lines represent the 95% equal-tailed credible intervals. In each instance, the studies included in estimating the diagnostic accuracy estimates are in grey points underlying the diamonds. From the study-specific grey points, there was substantial variation in both sensitivity and specificity between the studies and some studies had outlying values. It is also apparent that the number of tests evaluated differed among studies.

4.1 All available data (black diamonds)

The results presented are from the model assuming compound symmetry (see Supplementary material: Results1e.xlsx). This is because the model fitted the data better as reflected by lower WAIC, better mixing of chains and reduction in autocorrelation (reproducible trace plots not shown here) compared to the model assuming heterogeneous within-study variances (see supplementary material: Results1a.xlsx).

4.1.1 Triage of women with ASC-US to detect CIN2+

According to Figure 2, Linear Array was the most sensitive (0.92[0.86, 0.96]) but among the least specific (0.54 [0.43, 0.65]) tests while HPV Proofer was the least sensitive (0.68 [0.59, 0.76]) and the most specific (0.79 [0.73, 0.84]) test.

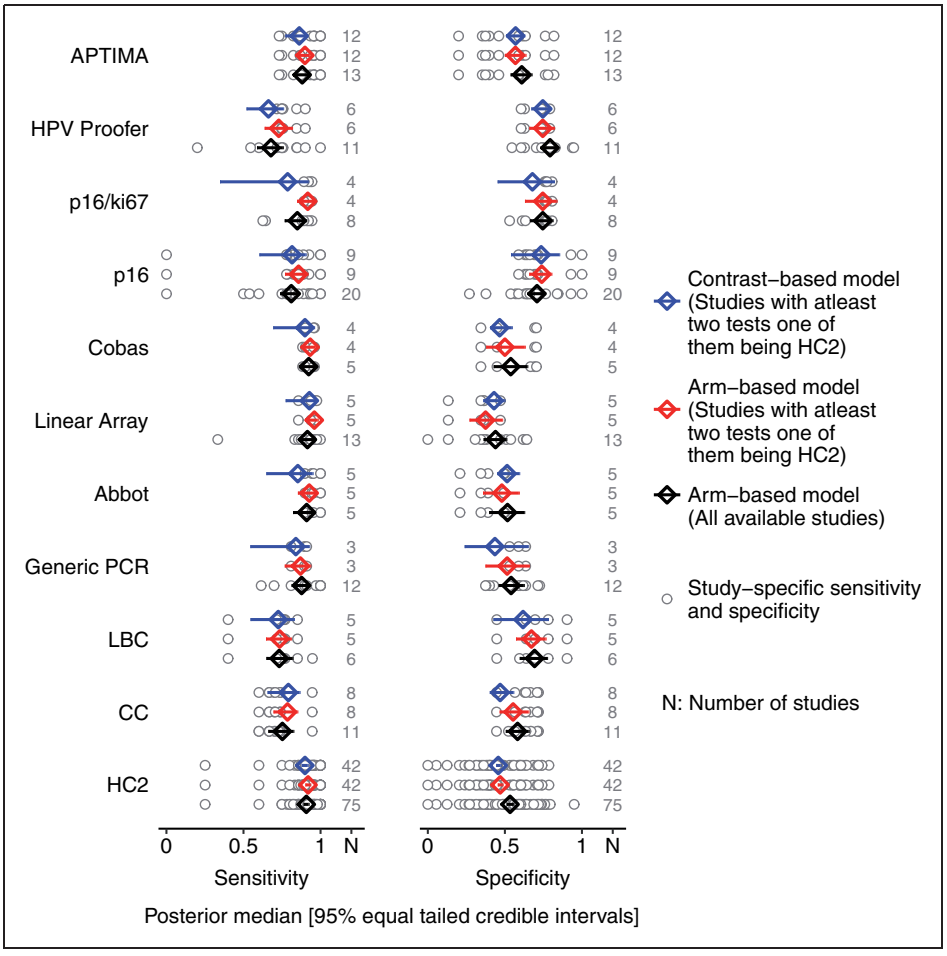


Figure 2. Cobas and Linear array were the most sensitive tests as estimated from all available and reduced data, respectively. HPV Proofer was the most specific test in detecting cervical intraepithelial neoplasia of grade 2 or worse in women with atypical squamous cervical cells of unspecified significance.

Both the diagnostic odds ratio and superiority index in the supplementary material (Results1e.xlsx) indicate that p16/Ki67 had the best discriminatory power with a sensitivity of 0.85 [0.77, 0.91] and specificity of 0.75 [0.66, 0.81]. Compared to HC2, LBC, p16 and HPV Proofer were less sensitive but more specific, while p16/Ki67 was equally as sensitive but more specific. All other tests had similar sensitivity and specificity as HC2 (see Table 1).

4.1.2 Triage of women with ASC-US to detect CIN3+

It can be seen in Figure 3 that Abbott RT PCR hrHPV was the most sensitive (0.98 [0.88, 1.00]) but among least specific (0.48 [0.35, 0.60]) tests. The diagnostic odds ratio and the superiority index (see supplementary material: Results1e.xlsx) indicate that p16/Ki67 had the best discriminatory power with sensitivity and specificity of 0.97 [0.84, 1.00] and 0.65 [0.53, 0.76], respectively. Relative to HC2, LBC, non-commercial PCR assays, p16 and HPV Proofer were less sensitive but more specific while CC, Abbott RT PCR hrHPV, Cobas-4800 were as sensitive and specific (see Table 2).

4.1.3 Triage of women with LSIL to detect CIN2+

Figure 4 and the absolute diagnostic estimates presented in the supplementary material (Results1e.xlsx) show that HC2 was the most sensitive (0.94 [0.93, 0.95]) test but among the least specific (0.29 [0.27, 0.31]) tests while HPV proofer was the least sensitive (0.64 [0.54, 0.73]) and the most specific (0.73 [0.67, 0.78]) test detecting CIN2+ in LSIL cytology. Both the diagnostic odds ratio and superiority index presented in the Supplementary material (Results1e.xlsx) indicate once more that p16/Ki67 had the best discriminatory power with an estimated sensitivity and specificity of 0.86 [0.79, 0.91] and 0.63 [0.57, 0.69].

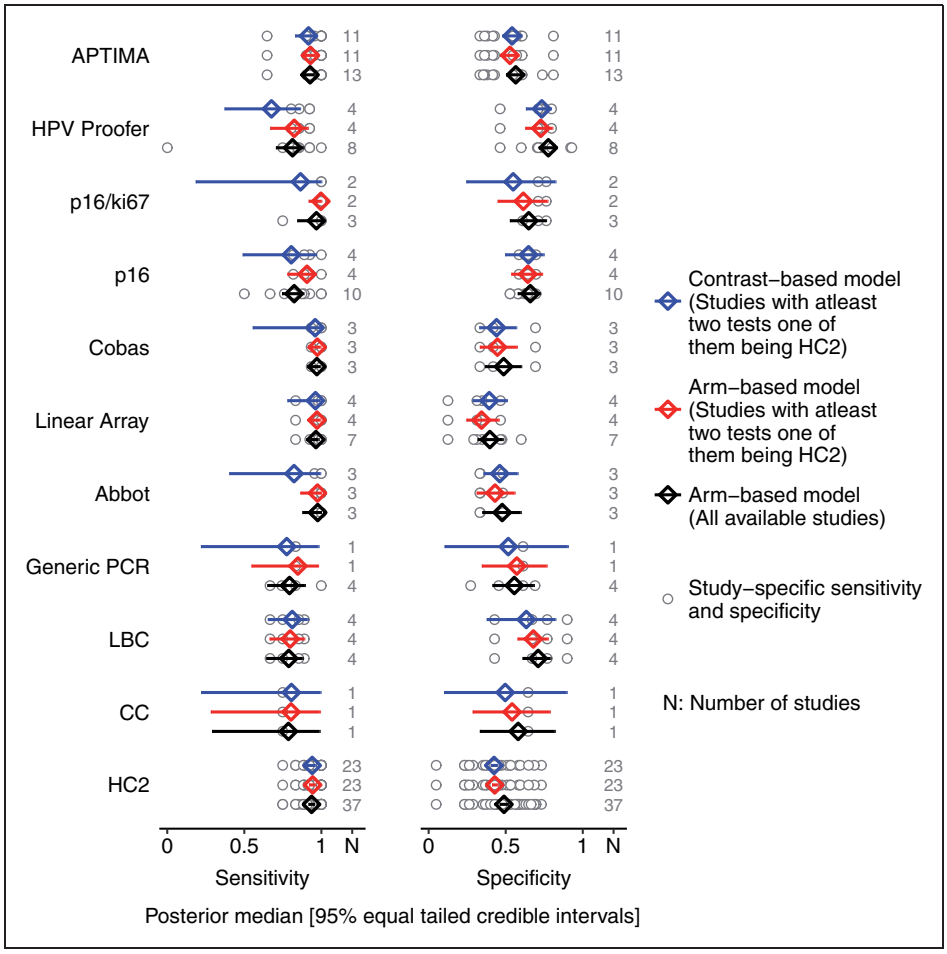


Figure 3. Abbot and Cobas were the most sensitive tests according to the arm- and contrast-based models, respectively, in cervical intraepithelial neoplasia of grade 3 or worse in women with atypical squamous cervical cells of unspecified significance.

4.1.4 Triage of women with LSIL to detect CIN3+

The forest plot presented in Figure 5 (see also supplementary material: Results1.xlsx) shows that Abbott RT PCR hrHPV and Linear Array were the most sensitive but among the least specific tests in detecting CIN3+ in women with LSIL. The diagnostic odds ratio indicate that Abbott RT PCR hrHPV had the best discriminatory power (sensitivity = 0.99 [0.95, 1.00], specificity = 0.28 [0.20, 0.38]). The superiority index indicates further that p16/Ki67 had equally best discriminatory power as Abbott RT PCR hrHPV with sensitivity and specificity of 0.95 [0.87, 0.98] and 0.47 [0.36, 0.57]. According to Tables 3 and 4, Abbott RT PCR hrHPV, Linear Array and Cobas-4800 were as sensitive and as specific while most of the rest of the tests were less sensitive but more specific as HC2 in detecting CIN2+ or CIN3+ in LSIL triage.

4.1.5 Variance components

The total variability in sensitivity (in the logit scale) from a compound symmetry working variance–covariance structure ranged from 0.23 [0.02, 0.76] (see supplementary material: Results1e.xlsx) in tests used to detect CIN3+ in ASC-US triage to 0.63 [0.40, 1.01] in tests used to detect CIN2+ in LSIL triage. The percentage of total variability in logit sensitivity attributable to between-study variability ranged from 42.52% [0.31%, 99.47%] in tests used to detect CIN3+ in LSIL triage to 81.38% [29.26%, 99.77%] in tests used to detect CIN2+ in ASC-US triage.

Similarly for logit specificity, the total variability ranged from 0.37 [0.25, 0.58] in tests used to detect CIN3+ in LSIL to 0.53 [0.40, 0.74] in tests detecting CIN2+ in ASC-US triage. Of the total variability in logit specificity, as low as 61.51% [33.91%, 78.40%] in tests used to detect CIN2+ in ASC-US triage and as high as 77.45% [59.42%,

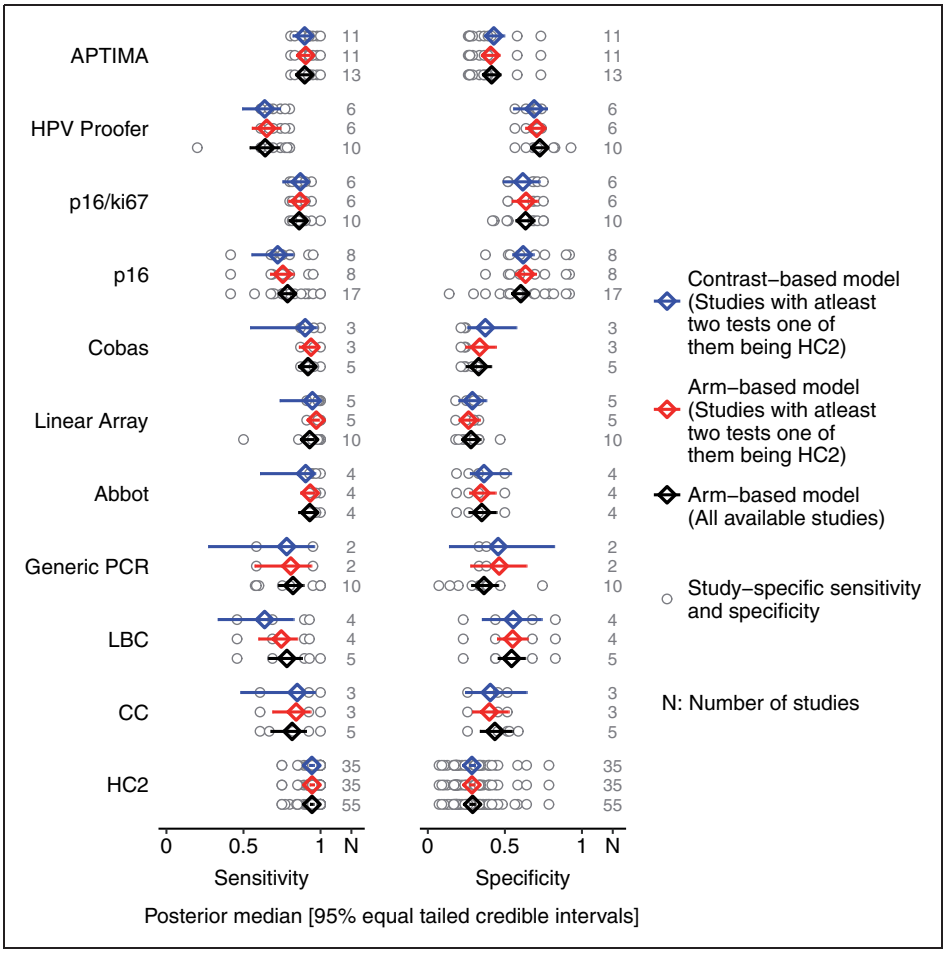


Figure 4. Abbot and Cobas were the most sensitive tests according to the arm- and contrast-based models, respectively, in cervical intraepithelial neoplasia of grade 2 or worse in women with low-grade squamous intraepithelial cervical lesions.

87.33%] in tests used to detect CIN2+ in LSIL triage were due to between-study heterogeneity. In other words, there was a stronger correlation between any two logit specificities in a given study than between any two logit sensitivities.

There was in general a stronger correlation between any two logit specificities in a given study than between any two logit sensitivities as indicated by larger between study variability of logit specificity.

The correlation between logit sensitivity and logit specificity was negative but not statistically significant except among tests used to detect CIN2+ in LSIL triage group ($\rho = -0.61$ [-0.88, -0.26]). The insignificant correlation parameters suggest absence of overall study effect in the respective data.

4.1.6 Sensitivity analysis

The sensitivity analysis did not highlight any particular change on the mean structure for different priors of the variance–covariance parameters (see supplementary material: Results1b-1d.xlsx). Based on the MCMC error sampling, the variance–covariance $\boldsymbol{\Sigma}$ was better sampled and less auto-correlated with LKJ and cauchy distributions.

4.2 AB versus CB model (black and red vs. blue)

The data were re-analysed to compare the estimates from the AB and CB models. Studies included in the re-analysis evaluated at least two tests with one of them being HC2 also set as the common comparator. The network plot of the studies included in the re-analyses is shown in Figure 6.

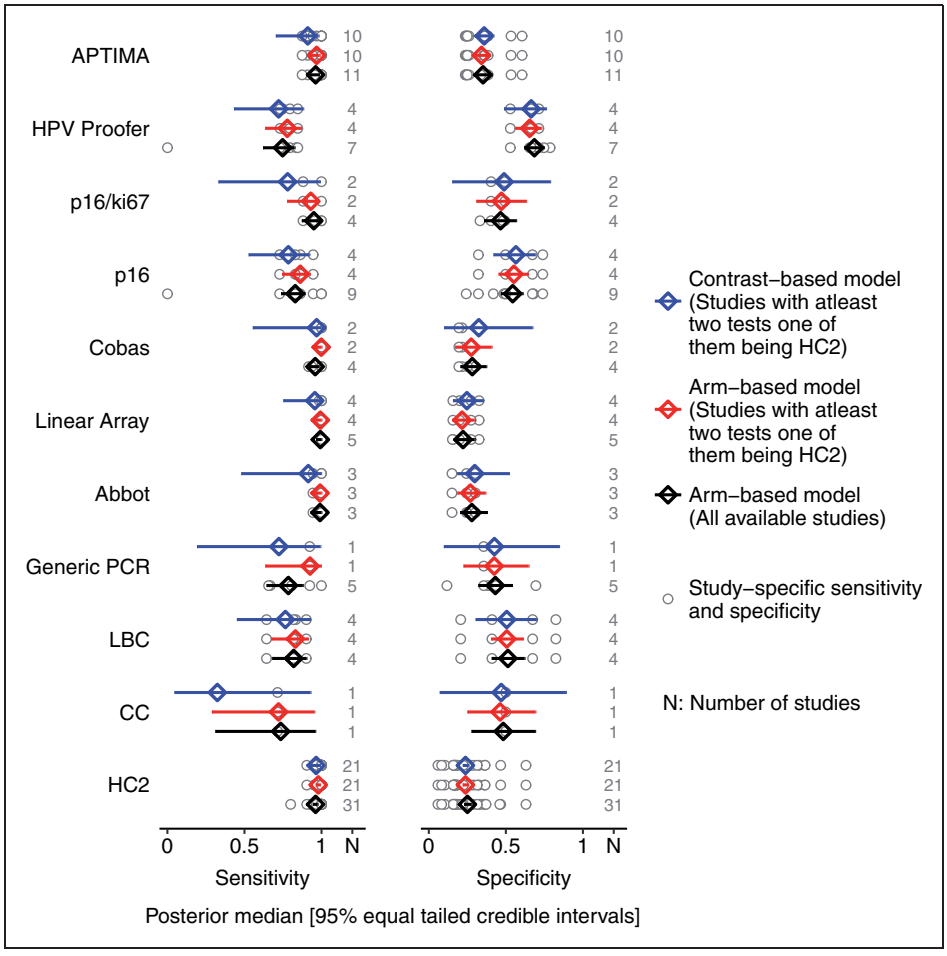


Figure 5. Linear array and HPV Proofer were the most sensitive and specific tests in detecting cervical intraepithelial neoplasia of grade 3 or worse in women with low-grade squamous intraepithelial cervical lesions.

Table 1. Posterior relative sensitivity and specificity and the corresponding 95% equal-tailed credible intervals of other tests relative to HC2 in detecting cervical intraepithelial neoplasia of grade 2 or worse in women with atypical squamous cervical cells of unspecified significance estimated by the arm-based model.

	Relative Sensitivity			Relative Specificity		
Test	Mean	Lower	Upper	Mean	Lower	Upper
HC2	1	1	1	1	1	1
Conventional cytology	0.83	0.73	0.91	1.09	0.94	1.23
Liquid-based cytology	0.81	0.71	0.90	1.30	1.11	1.45
Generic PCR assays	0.96	0.89	1.02	1.02	0.86	1.17
Abbott RT PCR hrHPV	1.00	0.91	1.06	0.97	0.75	1.18
Linear array	1.01	0.96	1.05	0.82	0.68	0.96
Cobas-4800	1.01	0.95	1.06	1.01	0.80	1.21
P16	0.89	0.81	0.95	1.33	1.20	1.45
P16/Ki67	0.93	0.85	1.00	1.39	1.23	1.54
HPV Proofer (mRNA)	0.74	0.65	0.83	1.48	1.36	1.59
APTIMA (mRNA)	0.97	0.91	1.01	1.14	1.00	1.27

Table 2. Posterior relative sensitivity and specificity and the corresponding 95% credible interval of other tests relative to HC2 in detecting cervical intraepithelial neoplasia of grade 3 or worse in women with atypical squamous cervical cells of unspecified significance estimated by the arm-based model.

	Relative Sensitivity			Relative Specificity		
Test	Mean	Lower	Upper	Mean	Lower	Upper
HC2	1	1	1	1	1	1
Conventional cytology	0.80	0.31	1.06	1.18	0.67	1.66
Liquid-based cytology	0.83	0.69	0.94	1.45	1.24	1.63
Generic PCR assays	0.84	0.69	0.96	1.13	0.83	1.41
Abbott RT PCR hrHPV	1.03	0.94	1.08	0.98	0.70	1.22
Linear array	1.03	0.99	1.06	0.82	0.64	1.00
Cobas-4800	1.03	0.97	1.07	0.99	0.75	1.23
P16	0.88	0.79	0.95	1.34	1.16	1.51
P16/Ki67	1.02	0.90	1.07	1.32	1.08	1.58
HPV Proofer (mRNA)	0.86	0.76	0.95	1.59	1.43	1.74
APTIMA (mRNA)	0.99	0.94	1.02	1.16	1.02	1.28

Table 3. Posterior relative sensitivity and specificity and the corresponding 95% equal-tailed credible interval of other tests relative to HC2 in detecting cervical intraepithelial neoplasia of grade 2 or worse in women with low-grade squamous intraepithelial cervical lesions triage as estimated by the arm-based model.

	Relative sensitivity			Relative specificity		
Test	Mean	Lower	Upper	Mean	Lower	Upper
HC2	1	1	1	1	1	1
Conventional cytology	0.86	0.72	0.96	1.50	1.15	1.91
Liquid-based cytology	0.82	0.70	0.93	1.87	1.54	2.22
Generic PCR assays	0.87	0.77	0.94	1.26	0.95	1.59
Abbott RT PCR hrHPV	0.98	0.91	1.03	1.21	0.92	1.54
Linear Array	0.98	0.93	1.02	0.97	0.77	1.20
Cobas-4800	0.97	0.90	1.02	1.14	0.85	1.45
P16	0.83	0.77	0.89	2.07	1.84	2.30
P16/Ki67	0.91	0.84	0.97	2.18	1.93	2.42
HPV Proofer (mRNA)	0.68	0.58	0.77	2.50	2.25	2.76
APTIMA (mRNA)	0.95	0.90	0.99	1.43	1.24	1.63

Table 4. Posterior relative sensitivity and specificity and the corresponding 95% credible interval of other tests relative to HC2 in detecting cervical intraepithelial neoplasia of grade 3 or worse in women with low-grade squamous intraepithelial cervical lesions as estimated by the arm-based model.

	Relative sensitivity			Relative specificity		
Test	Mean	Lower	Upper	Mean	Lower	Upper
HC2	1	1	1	1	1	1
Conventional cytology	0.73	0.32	1.00	1.93	1.10	2.78
Liquid-based cytology	0.85	0.70	0.94	2.05	1.62	2.48
Generic PCR assays	0.81	0.67	0.92	1.72	1.24	2.21
Abbott RT PCR hrHPV	1.03	0.99	1.05	1.12	0.79	1.49
Linear Array	1.03	1.00	1.05	0.90	0.66	1.19
Cobas-4800	0.99	0.93	1.03	1.13	0.81	1.52
P16	0.86	0.77	0.92	2.17	1.83	2.54
P16/Ki67	0.98	0.91	1.03	1.86	1.41	2.32
HPV Proofer(mRNA)	0.77	0.64	0.86	2.73	2.38	3.08
APTIMA(mRNA)	1.00	0.96	1.03	1.41	1.19	1.65

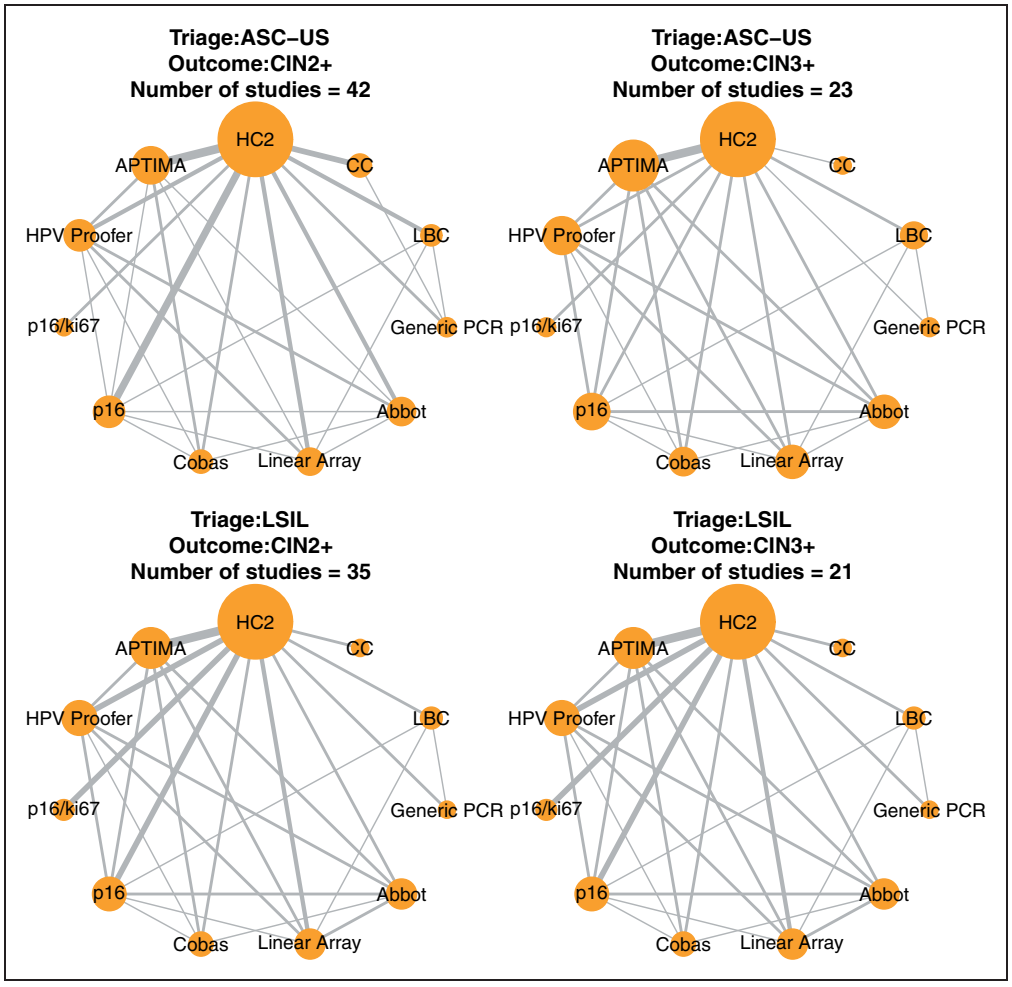


Figure 6. Network plot: Included studies evaluated at least two tests with HC2 as the common comparator.

The posterior estimates of the re-analysis are given in the Supplementary material. With a lower WAIC, the compound symmetry structure assumed in the AB model also fitted the reduced datasets more appropriately (see Supplementary material: Results2a.xlsx vs. Results2b.xlsx). The CB model with unstructured variance–covariance structure was unidentifiable. The block-diagonal variance–covariance structure modeled by the CB model fitted the reduced datasets better than the simple diagonal variance structure (see Supplementary material: Results3a.xlsx vs. Results3b.xlsx). Overall, the AB model had lower WAIC than the best CB model and therefore fitted the reduced datasets more appropriately. A graphical summary of the results from the second and third analysis are presented in Figures 2 to 5 and represented by the red and blue diamonds from the AB and CB models, respectively. Overall, there are discrepancies between the locations of the black, red and blue diamonds.

The location of the black diamonds is estimated from all available data, including studies evaluating single tests while the location of the red and blue diamonds is determined by studies evaluating at least two studies with one of them being HC2.

As a consequence of excluding some studies, the location of the black (full) and red (reduced datasets) diamonds are different though obtained using the same AB model. This illustrates the obvious fact that exclusion of studies could lead to different conclusions and recommendations.

The posterior median and the 95% equal-tailed credible intervals from the CB model are quite different from those of the AB model even with similar datasets. The wider credible intervals from the CB model could be attributed to loss of information by ignoring the correlation between the contrasts and as consequence the correlation between sensitivity and specificity. Moreover, the best fitting CB covariance structure was the block-diagonal of order K^2. The over-parameterisation of the covariance structure might also have contributed

to the inefficient estimation and potentially poor assessment of standard errors of the marginal means. Furthermore, difficulty in CB model identification is known to increase with number of tests of interest in the model.[4]

As a cascade effect, the ranking of the tests based on the DOR and the superiority also changes (see Supplementary material: Results1e.xlsx vs. Results2e.xlsx vs. Results3b.xlsx).

5 Discussion

In this paper, we propose a conceptually simple model to estimate sensitivity and specificity of multiple tests within a network meta-analysis framework analogous to a single-factor analysis of variance method with repeated measures.

The model is based on the assumption that all the tests were hypothetically used but missing at random. When the mechanism of missing data is not a crucial aspect of inference, models ignoring the missing value mechanism and only using the observed data as the proposed model does provide valid answers under a missing at random (MAR) process. In contrast to the CB model, the proposed AB model uses all available data in line with principle of intention-to-treat (ITT).[2] The missing 'unobservable' sensitivities and specificities are parameters that are estimated along with the other parameters in the model based on the exchangeability assumption. The cost, however, is that the model assumptions cannot be formally checked from the data under analysis.

When the data were never intended to be collected in the first place, the MAR assumption has been shown to hold as is the case in diagnostic studies where older tests become less used and new tests progressively more available with time.[35]

In the analysis, we included studies with at least one test. This is still acceptable because such studies still provide partial information allowing estimation of the mean and the variance–covariance parameters and only the study effects estimates might have larger standard errors.[30]

The proposed AB model allows for easy estimation of the marginal means and credible intervals for the intra-class correlation. Bayesian methods are known to be computationally intensive but with efficient sampling algorithms such as Hamilton Monte Carlo sampling implemented in Stan,[31] convergence to a stationary distribution is accelerated even with poor initial values. Furthermore, parallel chain processing greatly reduces computational time.

With the logit transformation, it is assumed that the transformed data are approximately normal with constant variance. For binary data as well as proportions, the mean and variance depend on the underlying probability. Therefore, any factor affecting the probability will change the mean and the variance. This implies that a linear model where the predictors affect the mean but assume a constant variance will not be adequate. Nonetheless, when the model for the mean is correct but the true distribution is not normal, the maximum likelihood (ML) estimates of the model parameters will be consistent but the standard errors will be incorrect.[36] An alternative to the logit transformation would be a variance stabilising angular transformation; however, the variance stabilising property of the transform depends on each n being large.[37]

The natural and optimal modelling approach would be to use the beta distribution. This was the motivation behind our work on copula-based bivariate beta distribution in meta-analysis of diagnostic data.[38,39] Our further research will focus on how different mean and correlation structures are accommodated and modelled using the beta-binomial distribution in network meta-analysis of diagnostic data.

There were discrepancies in identifying the best test between the DOR and the superiority index. While the range of values estimated by the two measures range from 0 to infinity, the DOR yields larger values than the superiority index. From the full dataset, the superiority index consistently identified p16/Ki67 as the best test in detecting cervical precancer with equivocal or mildly abnormal cervical cells. From the reduced data, the DOR identified tests with very low sensitivity but high specificity or vice-versa as the best and in disagreement with the superiority index. This illustrates that DOR cannot distinguish between tests with high sensitivity but low specificity or vice versa. In contrast, the superiority index gives more weight to tests performing relatively well on both diagnostic accuracy measures and less weight on tests performing poorly on both diagnostic measures or tests performing better on one measure but poorly on the other.[26] Nonetheless, both measures do not allow to prioritise one parameter which may be clinically appropriate. For example, presence of CIN2+ and especially CIN3+ indicates a considerable risk of developing cervical cancer and therefore cases should not be missed by a test. Consequently, higher test sensitivity has more clinical importance compared to specificity.

Incoherence or inconsistency within NMA is a major concern where for the same contrast, the direct and indirect evidence differ substantially. Lu and Ades,[40] Dias et al.[41] and Krahn et al. [42] explain how to visualise,

detect and handle inconsistencies. Since the AB model implicitly assumes consistency, the methods used to detect and quantify inconsistency in CB need not be used in the AB models.

For the AB models, Hong et al.[43] measure inconsistency by data-driven magnitude of bias, the discrepancy between observed and imputed treatment (test) effects while Piepho[44] classifies grouping of studies according to the set of tests included and introduces an interaction term: designs by test, to represent inconsistency.

From our viewpoint, inconsistency/incoherence is a form of heterogeneity between the studies which is often due to missing information in an outlying or influential study. In our model, the influence of the study on the mean is adequately captured by study-effects and the fact that the model hypothetically allows any two tests to be compared directly within each study makes inconsistency less an issue. That said, it is important to assess and identify influence of certain observations on the marginal mean. Detection of influential observations within the Bayesian framework is a computationally involved exercise and still an active research area.

This article does not consider individual-level data for which the model adaptation is automatic. Future research includes a study on impact of various aspects of data missingness on the robustness of the models.

Colposcopy, as used in the studies of our motivating datasets, is not a perfect standard. The proposed methodology does not take this into account and results might be somewhat biased and important differences between tests may be obscured.[45]

Approaches to minimise the bias due to the use of an imperfect gold standard include the application of latent class models,[4] studying sensitivity and specificity in different populations, use of an expert review panel to arrive at a less error-prone diagnosis[8] and framing the problem in terms of clinical outcomes rather than just accuracy.[45] A review of these approaches is presented by Reitsma et al.[46] The integration of such approaches within the AB network modelling is beyond the scope of this paper but is scheduled as future research.

Partial unbalanced verification of test-positive and test-negative subjects may yield substantial bias in the accuracy estimates. Overestimation of sensitivity and underestimation of specificity is the typical result when the majority of test-positives and only a small fraction of test-negatives are submitted to gold standard assessment. In the motivating dataset, partial verification was minimised in principle by restricting inclusion of studies with (nearly) complete assessment with the reference standard.[12,13] There are other methods to adjust for verification bias[8,47] when verification in all study subjects is not possible.

6 Conclusion

The proposed AB model contributes to the knowledge on methods used in systematic reviews of diagnostic data in presence of more than two competing tests. The AB model is more appealing than the CB model for meta-analyses of diagnostic studies because the model parameters permit a more straightforward interpretation and use all available data. Furthermore, more natural variance–covariance matrix structures can be easily accommodated.

Acknowledgements

We are grateful to the two anonymous reviewers for all their valuable comments that helped to improve the paper considerably.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: Nyaga received financial support from the Scientific Institute of Public Health (Brussels) through the OPSADAC project. Arbyn was supported by the COHEAHR project funded by the 7th Framework Programme of the European Commission (grant no. 603019). Aerts was supported by the IAP research network nr P7/06 of the Belgian Government (Belgian Science Policy).

Supplementary material

Supplementary material is available for this article online.

References

1. Higgins J and Whitehead A. Borrowing strength from external trials in a meta-analysis. *Stat Med* 1996; **15**: 2733–2749.
2. Fisher L, Dixon D, Herson J, et al. Intention to treat in clinical trials. In: Peace KE (ed.) *Statistical issues in drug research and development*. New York, NY: Marcel Dekker, 1989, pp.331–350.
3. Lumley T. Network meta-analysis for indirect treatment comparisons. *Stat Med* 2002; **21**: 2313–2324.
4. Menten J and Lesaffre E. A general framework for comparative bayesian meta-analysis of diagnostic studies. *BMC Med Res Methodol* 2015; **15**: 1.
5. Senn S. The many modes of meta. *Drug Inform J* 2000; **34**: 535–549.
6. Whitehead A. *Meta-analysis of controlled clinical trials*. Vol. 7, West Sussex, UK: John Wiley & Sons, 2002.
7. Piepho HP, Williams E and Madden L. The use of two-way linear mixed models in multitreatment meta-analysis. *Biometrics* 2012; **68**: 1269–1277.
8. Zhou XH, Obuchowski NA and McClish DK. *Design of diagnostic accuracy studies*. Hoboken, NJ: John Wiley & Sons, 2011.
9. Sinclair JC and Bracken MB. Clinically useful measures of effect in binary analyses of randomized trials. *J Clin Epidemiol* 1994; **47**: 881–889.
10. Davies HTO, Crombie IK and Tavakoli M. When can odds ratios mislead? *BMJ* 1998; **316**: 989–991.
11. Zhang J, Carlin BP, Neaton JD, et al. Network meta-analysis of randomized clinical trials: reporting the proper summaries. *Clin Trial* 2014; **11**: 246–262.
12. Arbyn M, Ronco G, Anttila A, et al. Evidence regarding human papillomavirus testing in secondary prevention of cervical cancer. *Vaccine* 2012; **30**: F88–F99.
13. Arbyn M, Roelens J, Simoens C, et al. Human papillomavirus testing versus repeat cytology for triage of minor cytological cervical lesions. *Cochrane Database Syst Rev* 2013; **3**: CD008054.
14. Arbyn M, Roelens J, Cuschieri K, et al. The aptima hpv assay versus the hybrid capture 2 test in triage of women with ascus or lsil cervical cytology: a meta-analysis of the diagnostic accuracy. *Int J Cancer* 2013; **132**: 101–108.
15. Roelens J, Reuschenbach M, von Knebel Doeberitz M, et al. p16ink4a immunocytochemistry versus human papillomavirus testing for triage of women with minor cytologic abnormalities. *Cancer Cytopathol* 2012; **120**: 294–307.
16. Verdoodt F, Szarewski A, Halfon P, et al. Triage of women with minor abnormal cervical cytology: meta-analysis of the accuracy of an assay targeting messenger ribonucleic acid of 5 high-risk human papillomavirus types. *Cancer Cytopathol* 2013; **121**: 675–687.
17. Bosch F, Lorincz A, Munoz N, et al. The causal relation between human papillomavirus and cervical cancer. *J Clin Pathol* 2002; **55**: 244–265.
18. Arbyn M, Ronco G, Cuzick J, et al. How to evaluate emerging technologies in cervical cancer screening? *Int J Cancer* 2009; **125**: 2489–2496.
19. Daniels MJ and Pourahmadi M. Modeling covariance matrices via partial autocorrelations. *J Multivariate Anal* 2009; **100**: 2352–2363.
20. Reitsma JB, Glas AS, Rutjes AW, et al. Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *J Clin Epidemiol* 2005; **58**: 982–990.
21. Chu H and Cole SR. Bivariate meta-analysis of sensitivity and specificity with sparse data: a generalized linear mixed model approach. *J Clin Epidemiol* 2006; **59**: 1331–1332.
22. Altham PM. Improving the precision of estimation by fitting a model. *J Royal Stat Soc Ser B* 1984; **46**: 118–119.
23. Verbeke G and Molenberghs G. Inference for the variance components. In: Bickel P, Diggle P, Fienberg S, et al. (eds) *Linear mixed models for longitudinal data*. New York, NY: Springer Series in Statistics, 2002, p.64.
24. Watanabe S. Asymptotic equivalence of bayes cross validation and widely applicable information criterion in singular learning theory. *J Mach Learn Res* 2010; **11**: 3571–3594.
25. Glas AS, Lijmer JG, Prins MH, et al. The diagnostic odds ratio: a single indicator of test performance. *J Clin Epidemiol* 2003; **56**: 1129–1135.
26. Deutsch R, Mindt MR, Xu R, et al. Quantifying relative superiority among many binary-valued diagnostic tests in the presence of a gold standard. *J Data Sci* 2009; **7**: 161–177.
27. Rubin DB. Inference and missing data. *Biometrika* 1976; **63**: 581–592.
28. Little RJ and Rubin DB. *Statistical analysis with missing data*. New York: John Wiley & Sons, 1987.
29. Lewandowski D, Kurowicka D and Joe H. Generating random correlation matrices based on vines and extended onion method. *J Multivariate Anal* 2009; **100**: 1989–2001.
30. Gelman A and Hill J. Multilevel linear models: varying slopes, non-nested models, and other complexities. In: Alvarez RM, Beck NL and Wu LL (eds) *Data analysis using regression and multilevel/hierarchical models*. Cambridge, UK: Cambridge University Press, 2006, p.286.
31. Carpenter B, Gelman A and Hoffman M. Stan: a probabilistic programming language. *Journal of Statistical Software* 2017, DOI: 10.18637/jss.v076.i01
32. Hoffman MD and Gelman A. The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. *J Mach Learn Res* 2014; **15**: 1593–1623.

33. Team RC. *A Language and environment for statistical computing*. *R Foundation for Statistical Computing*. Vienna, Austria, http://www.R-project.org/ (2015, accessed 1 January 2015).
34. Team SD. *Stan: A C++ library for probability and sampling, Version 2.9.0*, http://mc-stan.org/ (2015, accessed 1 January 2015).
35. Schafer JL and Graham JW. Missing data: our view of the state of the art. *Psychol Meth* 2002; **7**: 147.
36. Agresti A. Generalized linear models for counts. In: Balding DJ, Cressie NAC, Fitzmaurice GM, Goldstein H, et al. (eds) *Categorical data analysis*, 2nd ed. New York, NY: John Wiley & Sons, 2002, p.131.
37. Crowder MJ. Beta-binomial anova for proportions. *Appl Stat* 1978; **27**: 34–37.
38. Nyaga V. *CopulaDTA: Copula based bivariate beta-binomial model for diagnostic test accuracy studies*, R package version 0.0.2, https://cran.r-project.org/package=CopulaDTA (2015, accessed 1 January 2015).
39. Nyaga V, Aerts M and Arbyn M. Marginal models for meta-analysis of diagnostic accuracy studies in frequentist and Bayesian framework using rstan and CopulaREMADA. *Archive Public Health* 2015; **73**(Suppl 1): O4.
40. Lu G and Ades A. Combination of direct and indirect evidence in mixed treatment comparisons. *Stat Med* 2004; **23**: 3105–3124.
41. Dias S, Welton N, Caldwell D, et al. Checking consistency in mixed treatment comparison meta-analysis. *Stat Med* 2010; **29**: 932–944.
42. Krahn U, Binder H and König J. Visualizing inconsistency in network meta-analysis by independent path decomposition. *BMC Med Res Method* 2014; **14**: 131.
43. Hong H, Chu H, Zhang J, et al. A bayesian missing data framework for generalized multiple outcome mixed treatment comparisons. *Res Synth Meth* 2015; **7**: 6–22.
44. Piepho HP. Network-meta analysis made easy: detection of inconsistency using factorial analysis-of-variance models. *BMC Med Res Methodol* 2014; **14**: 61.
45. Valenstein PN. Evaluating diagnostic tests with imperfect standards. *Am J Clin Pathol* 1990; **93**: 252–258.
46. Reitsma JB, Rutjes AW, Khan KS, et al. A review of solutions for diagnostic accuracy studies with an imperfect or missing reference standard. *J Clin Epidemiol* 2009; **62**: 797–806.
47. Arbyn M, Ronco G, Cuzick J, et al. How to evaluate emerging technologies in cervical cancer screening? *Int J Cancer* 2009; **125**: 2489–2496.